

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Full Stack Development Using Mern
Practical - V

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Class: TYIT	Batch: 2
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5.React Routing and Data Handling :

a. Install and configure React Router in the React application.

```
PS C:\Users\info\OneDrive\Desktop\Mern Stack Practical\Practical5> npm install react-router-dom
added 4 packages, and audited 140 packages in 2s
32 packages are looking for funding
run `npm fund` for details
found 0 vulnerabilities
```

b. Create multiple pages and implement navigation between them.

CODE: -

Home.jsx :

```
function Home () {
  return (
    <div>
      <h1> Home page </h1>
      <p> Welcome to our React Application . </p>
    </div>
  );
}
export default Home ;
```

Contact.jsx :

```
function Contact ()
{
  return (
    <div>

      <h1> Content page </h1>
      <p> Email : khushipatelmvlu@gmail.com . </p>
    </div>
  );
}
export default Contact;
```

About.jsx :

```
function About ()
{
  return (
```

```

    <div>

      <h1> Home page </h1>
      <p> This page contains information about our application </p>
    </div>

  );
}
export default About;

```

App.jsx :

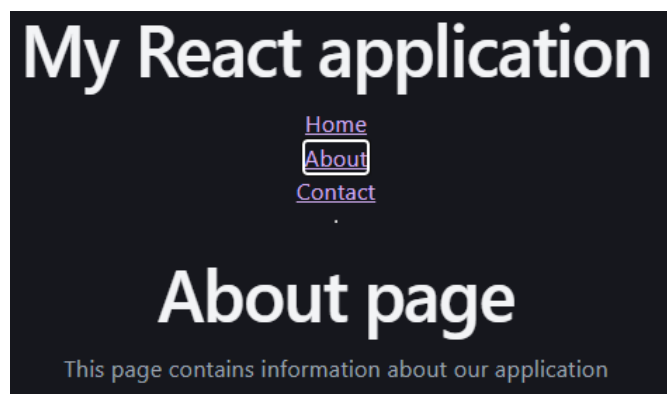
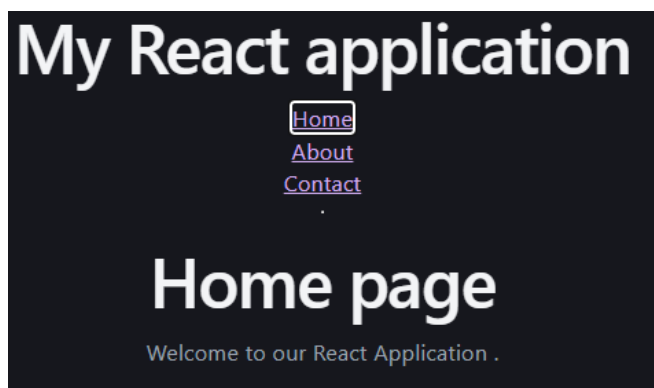
```

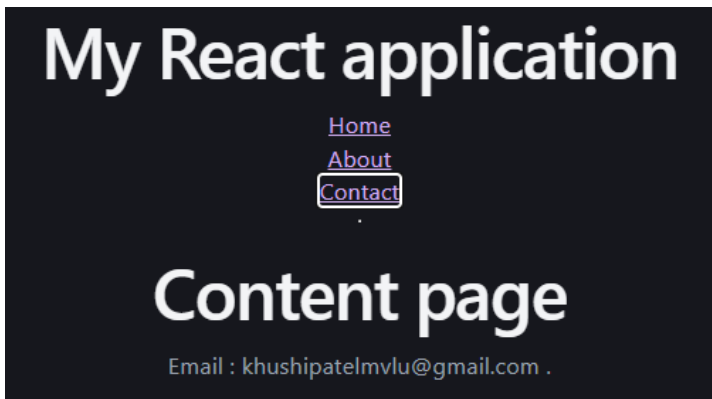
import {BrowserRouter , Routes, Route, Link } from "react-router-dom";
import Home from './pages/Home';
import About from './pages/About';
import Contact from './pages/Contact';
function App() {
  return (
    <BrowserRouter>
      <h1>My React application</h1>
      <nav>
        <Link to="/">Home</Link>
        <br/>
        <Link to="/About">About</Link>
        <br/>
        <Link to="/Contact">Contact</Link>
      </nav>
      <hr/>
      <Routes>
        <Route path="/" element={ <Home/> }/>
        <Route path="/about" element={ <About/> }/>
        <Route path="/Contact" element={ <Contact/> }/>
      </Routes>
    </BrowserRouter>
  );
}

```

export default App;

OUTPUT:-





c. Use Link and Routes to move between pages.

CODE: -

Profile.jsx :

```
function Profile()
{
  return (
    <div>
      <h1>Student Profile</h1>
      <hr />

      <p>Name: Khushii Patel</p>
      <p>Course: B.Sc. IT</p>
      <p>CGPA: 8.4</p>
      <p>Skills: JavaScript, React, Node.js</p>
    </div>
  );
}
```

export default Profile;

Dashboard.jsx :

```
function Dashboard() {
  return (
    <div>
      <h1>Placement Dashboard</h1>
      <p>Welcome to the Student Placement Portal.</p>
      <hr />

      <h2>Placement Summary</h2>
      <p>Total Companies: 25</p>
      <p>Students Placed: 120</p>
      <p>Highest Package: ₹12 LPA</p>
    </div>
  );
}
```

export default Dashboard;

Jobs.jsx :

```
function Jobs() {
  return (
    <div>
      <h1>Job Opportunities</h1>
```

```

    <hr />

    <h2>Software Developer</h2>
    <p>Company: ABC Technologies</p>
    <p>Location: Mumbai</p>
    <p>Package: ₹6 LPA</p>

    <h2>Web Developer</h2>
    <p>Company: XYZ Solutions</p>
    <p>Location: Pune</p>
    <p>Package: ₹5 LPA</p>

    <h2>Data Analyst</h2>
    <p>Company: Tech Solutions</p>
    <p>Location: Bangalore</p>
    <p>Package: ₹7 LPA</p>
  </div>
);
}

export default Jobs;

Statistics.jsx :
function Statistics() {
  return (
    <div>
      <h1>Placement Statistics Page</h1>
      <p>This is the statistics page of our application.</p>
      <hr/>
      <h2>Statistics Details</h2>
      <p>Total Students: 100</p>
      <p>Placed Students: 50</p>
      <p>Unplaced Students: 50</p>
      <p>Average Package: 8 LPA</p>
    </div>
  )
}
export default Statistics;
import {BrowserRouter , Routes, Route, Link } from "react-router-dom";
import Dashboard from './Pages/Dashboard';
import Profile from './Pages/Profile';
import Jobs from './Pages/Jobs';
import Statistics from './Pages/Statistics';

```

```

App.jsx :
function App() {
  return (
    <BrowserRouter>
      <h1>M.V.L.U Placement Portal</h1>
      <nav>
        <Link to="/">Dashboard</Link>
        <br/>
        <Link to="/Profile">Profile</Link>
        <br/>

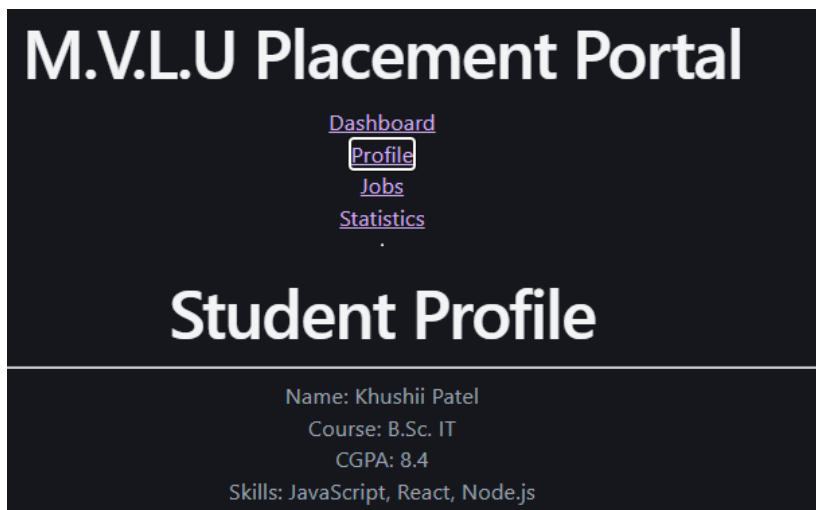
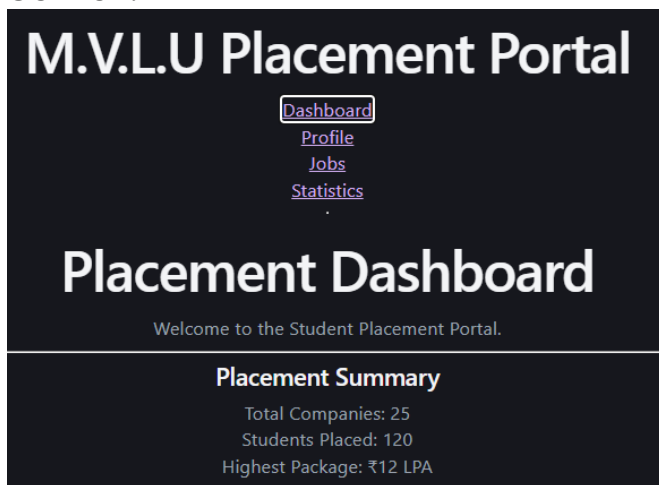
```

```

<Link to="/Jobs">Jobs</Link>
<br/>
<Link to="/Statistics">Statistics</Link>
<br/>
</nav>
<hr/>
<Routes>
  <Route path="/" element={<Dashboard/>}/>
  <Route path="/Profile" element={<Profile/>}/>
  <Route path="/Jobs" element={<Jobs/>}/>
  <Route path="/Statistics" element={<Statistics/>}/>
</Routes>
</BrowserRouter>
);
}
export default App;

```

OUTPUT:-



M.V.L.U Placement Portal

[Dashboard](#)
[Profile](#)
[Jobs](#)
[Statistics](#)

Job Opportunities

Software Developer

Company: ABC Technologies

Location: Mumbai

Package: ₹6 LPA

Web Developer

Company: XYZ Solutions

Location: Pune

Package: ₹5 LPA

Data Analyst

Company: Tech Solutions

Location: Bangalore

Package: ₹7 LPA

M.V.L.U Placement Portal

[Dashboard](#)
[Profile](#)
[Jobs](#)
[Statistics](#)

Placement Statistics Page

This is the statistics page of our application.

Statistics Details

Total Students: 100

Placed Students: 50

Unplaced Students: 50

Average Package: 8 LPA

d) Fetch sample data from a public API using Axios.

CODE:-

```
import axios from "axios";
import { useEffect, useState } from "react";
function App() {
  const [products, setProducts] = useState([]);
  useEffect(() => {
    axios
      .get("https://dummyjson.com/products")
      .then((response) => {
        setProducts(response.data.products);
      })
      .catch((error) => {
        console.log(error);
      });
  });
}
```

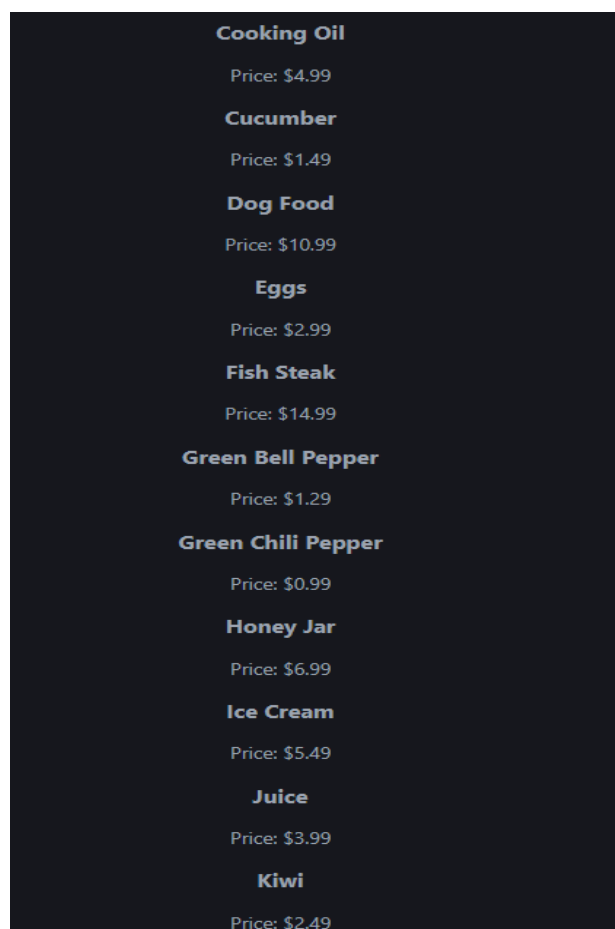
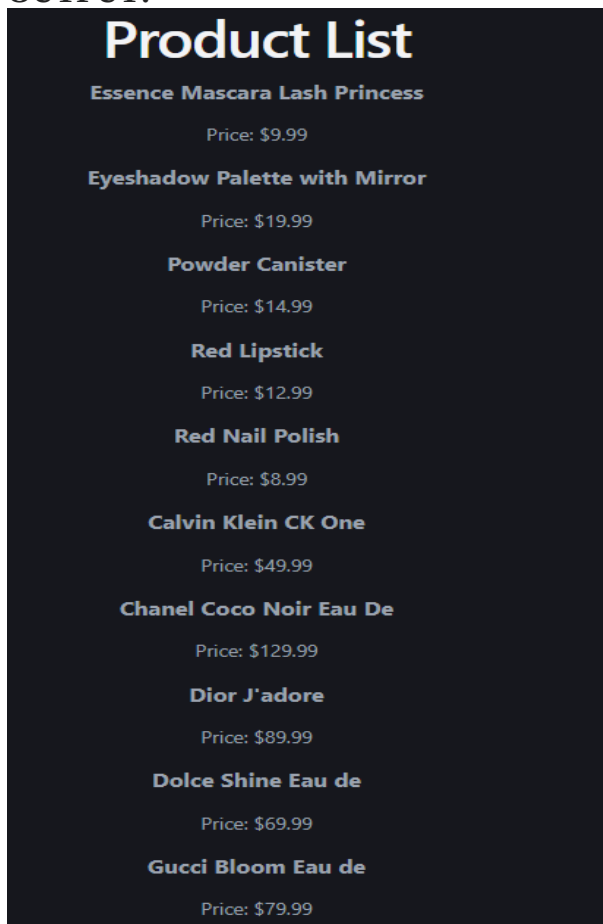
```

    });
  }, []);
  return (
    <div>
      <h1>Product List</h1>

      {products.map((product) => (
        <div key={product.id}>
          <h3>{product.title}</h3>
          <p>Price: ${product.price}</p>
        </div>
      ))}
    </div>
  );
}
export default App;

```

OUTPUT:-



e) Display fetched data in table or list format.

CODE :-

```

import axios from "axios";
import { useEffect, useState } from "react";

function App() {
  const [products, setProducts] = useState([]);

  useEffect(() => {
    axios
      .get("https://dummyjson.com/products")

```

```

.then((response) => {
  setProducts(response.data.products);
});
}, []);

return (
  <div>
    <h1>Product List</h1>
    <hr />

    <table border="1" cellPadding="10">
      <thead>
        <tr>
          <th>ID</th>
          <th>Product Name</th>
          <th>Category</th>
          <th>Price</th>
          <th>Rating</th>
          <th>Stock</th>
        </tr>
      </thead>

      <tbody>
        {products.map((product) => (
          <tr key={product.id}>
            <td>{product.id}</td>
            <td>{product.title}</td>
            <td>{product.category}</td>
            <td>${product.price}</td>
            <td>{product.rating}</td>
            <td>{product.stock}</td>
          </tr>
        ))}
      </tbody>
    </table>
  </div>
);
}

```

export default App;

OUTPUT:-

Product List					
ID	Product Name	Category	Price	Rating	Stock
1	Essence Mascara Lash Princess	beauty	\$9.99	2.56	99
2	Eyeshadow Palette with Mirror	beauty	\$19.99	2.86	34
3	Powder Canister	beauty	\$14.99	4.64	89
4	Red Lipstick	beauty	\$12.99	4.36	91
5	Red Nail Polish	beauty	\$8.99	4.32	79

6	Calvin Klein CK One	fragrances	\$49.99	4.37	29
7	Chanel Coco Noir Eau De	fragrances	\$129.99	4.26	58
8	Dior J'adore	fragrances	\$89.99	3.8	98
9	Dolce Shine Eau de	fragrances	\$69.99	3.96	4
10	Gucci Bloom Eau de	fragrances	\$79.99	2.74	91

11	Annibale Colombo Bed	furniture	\$1899.99	4.77	88
12	Annibale Colombo Sofa	furniture	\$2499.99	3.92	60
13	Bedside Table African Cherry	furniture	\$299.99	2.87	64
14	Knoll Saarinen Executive Conference Chair	furniture	\$499.99	4.88	26
15	Wooden Bathroom Sink With Mirror	furniture	\$799.99	3.59	7
16	Apple	groceries	\$1.99	4.19	8
17	Beef Steak	groceries	\$12.99	4.47	86
18	Cat Food	groceries	\$8.99	3.13	46
19	Chicken Meat	groceries	\$9.99	3.19	97
20	Cooking Oil	groceries	\$4.99	4.8	10
21	Cucumber	groceries	\$1.49	4.07	84
22	Dog Food	groceries	\$10.99	4.55	71
23	Eggs	groceries	\$2.99	2.53	9
24	Fish Steak	groceries	\$14.99	3.78	74
25	Green Bell Pepper	groceries	\$1.29	3.25	33

f) Implement search or filter functionality on displayed data.

CODE:-

```
import axios from "axios";
import { useEffect, useState } from "react";
function App() {
  const [products, setProducts] = useState([]);
  const [search, setSearch] = useState("");
  useEffect(() => {
    axios
      .get("https://dummyjson.com/products")
      .then((response) => {
        setProducts(response.data.products);
      });
  }, []);
  const filteredProducts = products.filter((product) =>
    product.title.toLowerCase().includes(search.toLowerCase())
  );
  return (
    <div>
      <h1>Product List</h1>
      <input
        type="text"
        placeholder="Search product..."
      />
    </div>
  );
}
```

```

    value={search}
    onChange={(e) => setSearch(e.target.value)}
  />
<hr />
<table border="1" cellPadding="10">
  <thead>
    <tr>
      <th>ID</th>
      <th>Product Name</th>
      <th>Category</th>
      <th>Price</th>
      <th>Rating</th>
      <th>Stock</th>
    </tr>
  </thead>
  <tbody>
    {filteredProducts.map((product) => (
      <tr key={product.id}>
        <td>{product.id}</td>
        <td>{product.title}</td>
        <td>{product.category}</td>
        <td>${product.price}</td>
        <td>{product.rating}</td>
        <td>{product.stock}</td>
      </tr>
    ))}
  </tbody>
</table>
</div>
);
}

```

export default App;

OUTPUT:-

Product List

ID	Product Name	Category	Price	Rating	Stock
10	Gucci Bloom Eau de	fragrances	\$79.99	2.74	91

Search box